

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- Overview of air quality models
- Expectation
 - Understand different types of air quality models

To start with.....

- What is a model?
 - A representation of a system
 - To investigate its properties and/or perform predictions
 - Vary substantially in complexity
- Why we need model in air quality field?
 - Measurements are expensive
 - Measurements are limited, spatially and temporally
 - Need to answer “what if” questions

To start with.....

- *“all models are wrong”, but useful*
- There are always errors and biases in model outputs
- Caution must be taken to evaluate model outputs, and to identify and correct biases and errors

To start with.....

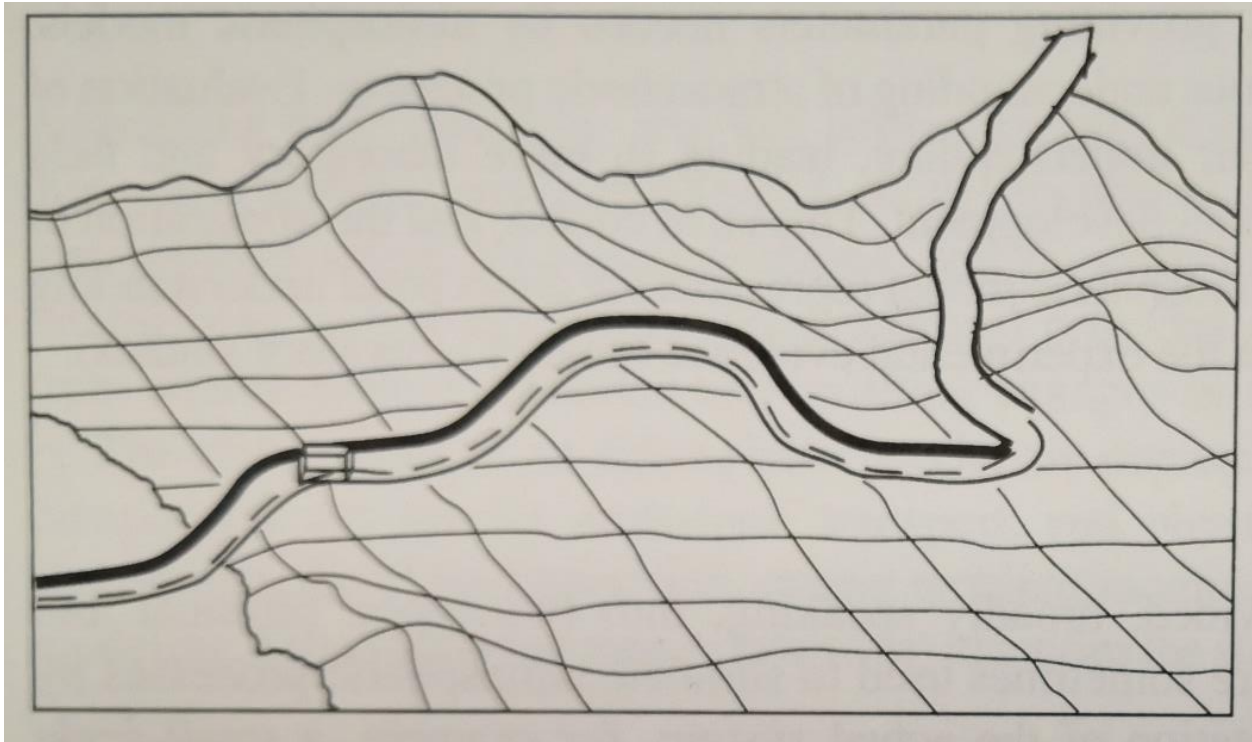
- Mean to, mechanistically, mathematically approximate actual physical chemical process in the atmosphere
 - Not a strict definition
- Use model to investigate the fate and transport of chemical species in the atmosphere
 - Note that this is for air quality purposes
 - Many advanced models include more than just atmosphere, and are capable of more purposes

Types of air quality models

- Spatial scale
 - Micro (hundreds of meters), urban (hundreds of km), regional (thousands of km), continental (several thousands of km), global
- Temporal scale
 - Short-term (up to dozens of days), long-term (years)
- Characterization of transport
 - Eulerian vs Lagrangian
- Dimensions
 - Zero (box model), 1D (column), 2D, 3D
- Target problems
 - Photochemical, acid deposition, aerosol, source apportionment, street canyon etc
- Mechanism
 - Dispersion, Chemical Transport Model (CTM), Computational Fluid Dynamics (CFD) etc

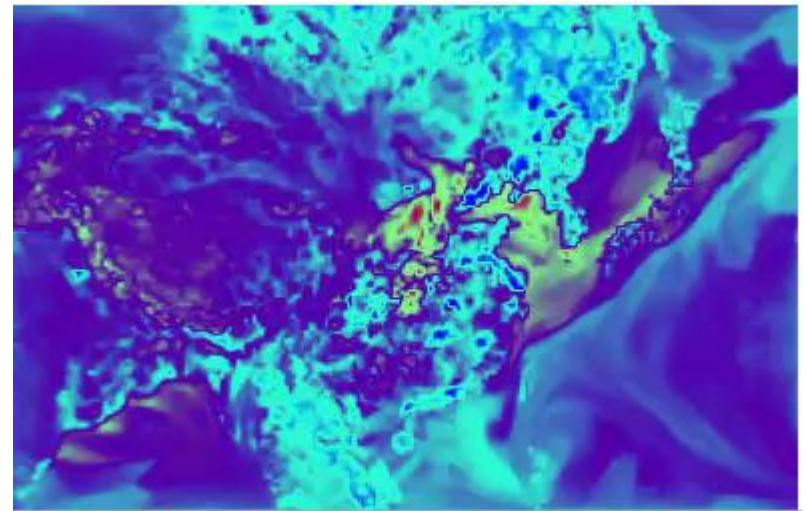
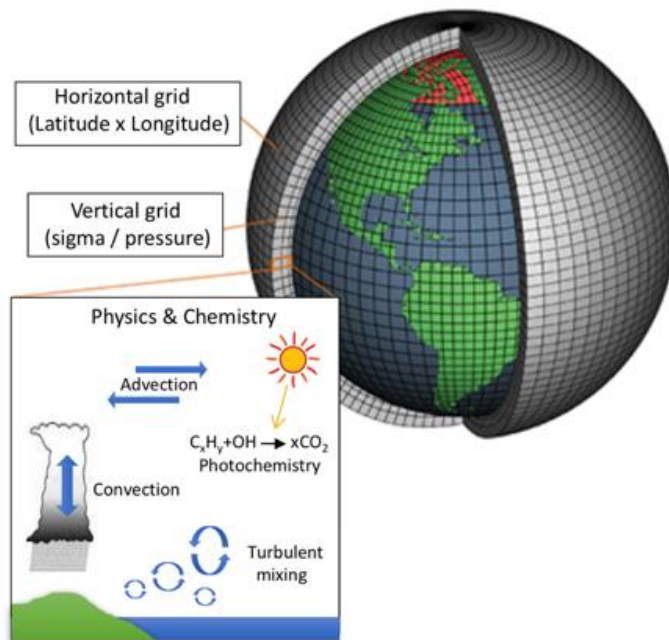
Lagrangian vs Eulerian approach

- Lagrangian: Focus on one 'piece' of pollutant, track it through space and time (stalker)



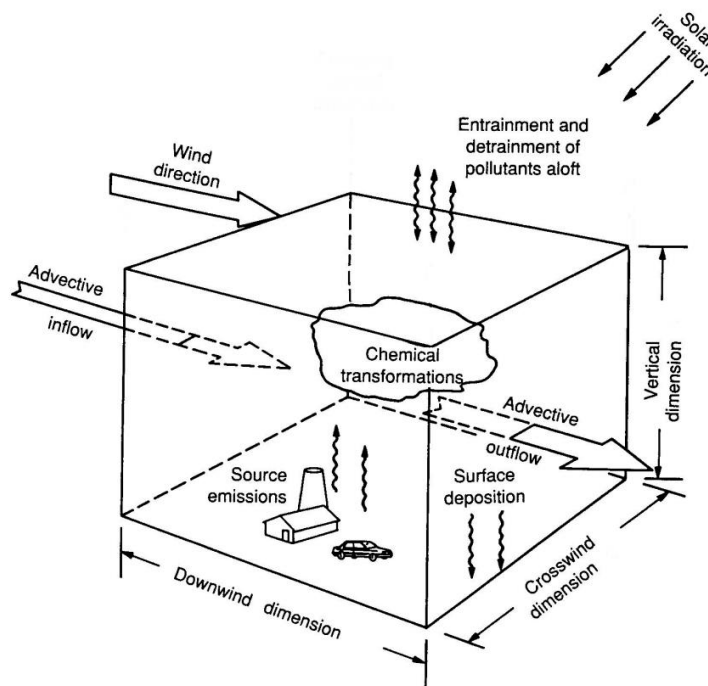
Lagrangian vs Eulerian approach

- Eulerian: Focus on the same area, estimate spatial concentrations distributions at each time step (snapshot)



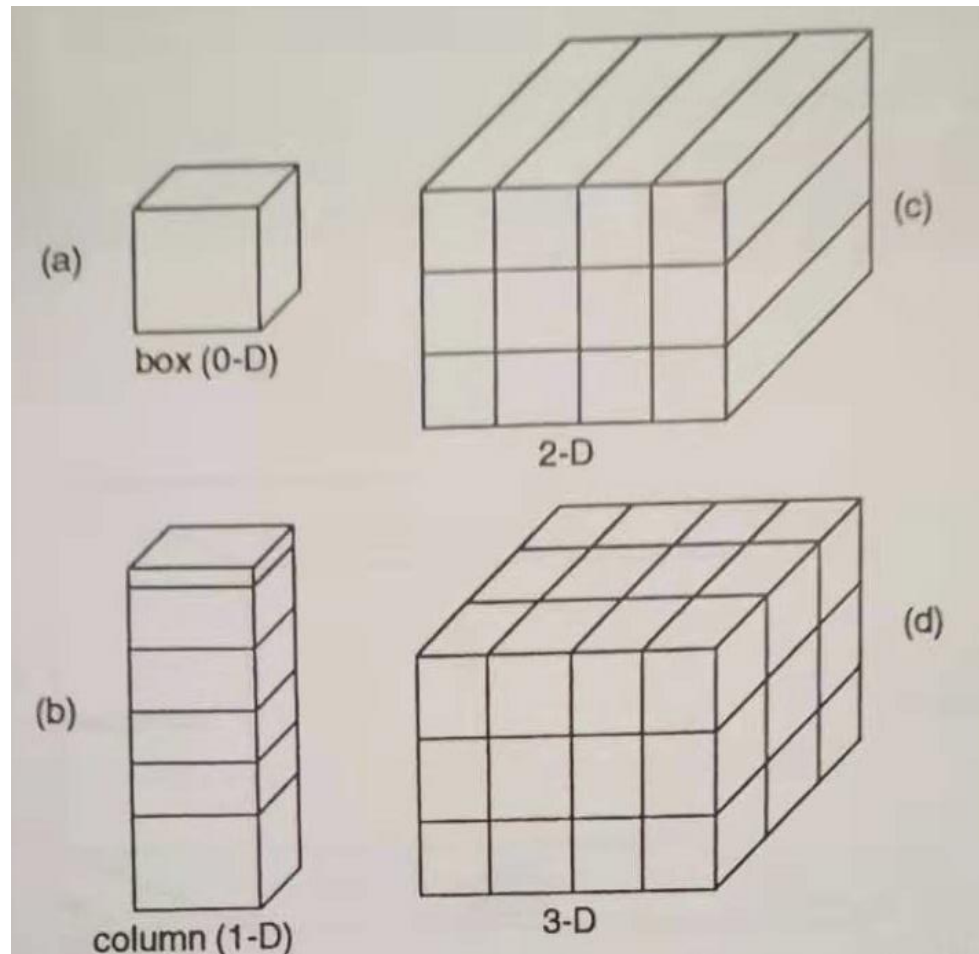
Box model

- Mass balance models, or 0-D models
 - Can be the simplest
 - Can be very complex (e.g. BOXMOX model)
- Treat entire study domain as one “box”, assume instantaneous mix inside the box
- Still widely used to study atmospheric chemistry



An example of a complex box model

Model dimensions

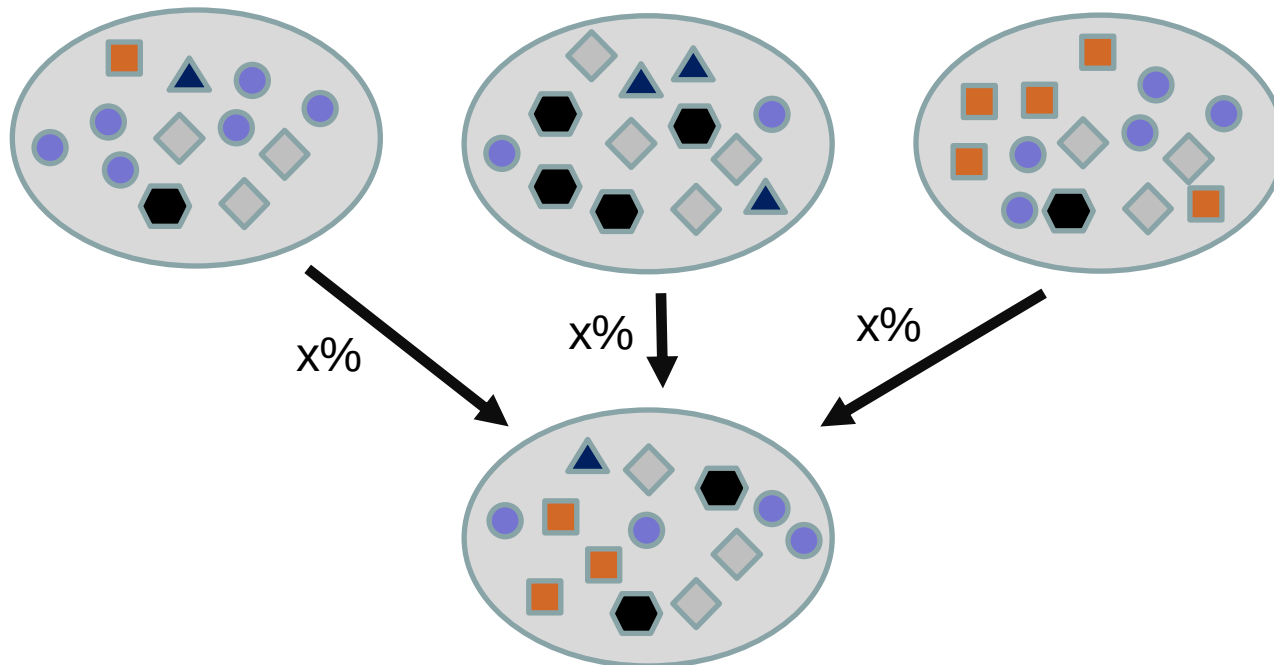


Target problems

- Photochemical models
 - Detailed gas-phase photochemical reactions
 - No aqueous and aerosol phase
- Acid deposition models
 - Focus on SO_2 & NO_x processes
 - Gas & aqueous & aerosol chemistry
- Aerosol models
 - Physical and chemical process of aerosol chemistry

Source appointment models

- Aka: receptor models
- “Identifying and quantifying the sources of air pollutants at a receptor location”
- Example: Chemical mass balance (CMB), Positive matrix factorization (PMF)



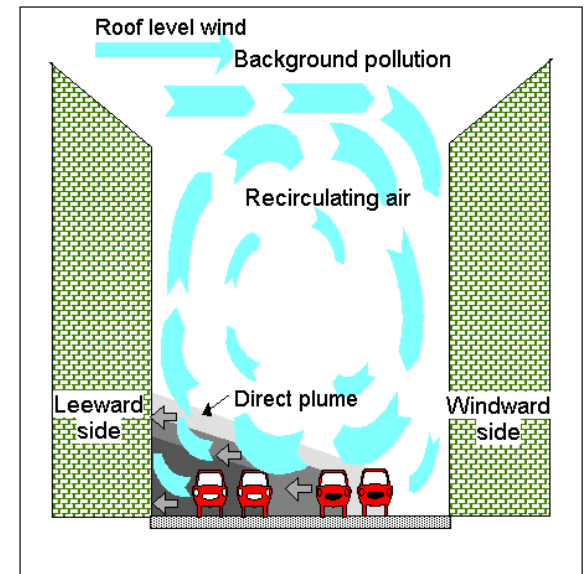
Receptor models

$$C_i = \sum_{j=1}^m \bar{f}_{ij} \cdot \bar{S}_j + e_i$$

- Require speciated (usually metals) particulate matter composition at the source(s) of interest
- **CMB**
 - Source profiles (f_{ij}) are known
 - Measured from sources
 - Solve the MBE using least squares method
 - About 8 equations in 20 unknowns
 - Can be done on each sample
- **PMF**
 - Both source profiles (f_{ij}) and source contributions (S_j) are unknown
 - Solve the MBE using constrained, weighted, least-squares, factor analytic approach.
 - Many samples required
 - A few hundred unknowns

Street canyon models

- Designed specifically for street canyons
- Estimation vertical concentration distribution in street canyons
 - Utilizing a wide range of approaches from simple regression to CFD modeling

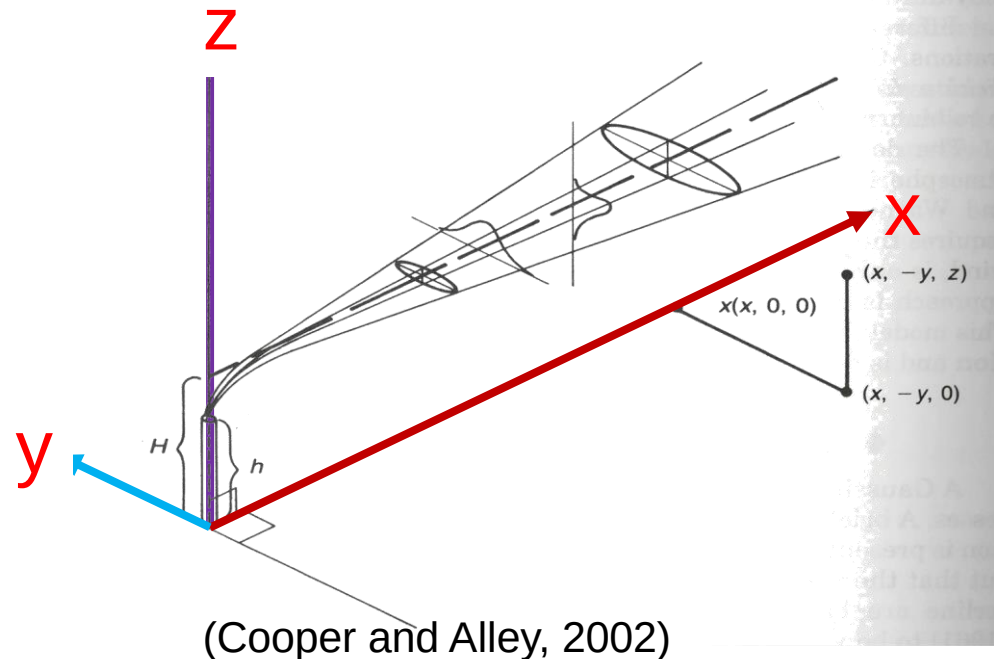


Gaussian dispersion models

- Steady-state vs non-steady-state models
- Steady-state
 - Assume homogeneous meteorological fields
 - Usually no “memory” effects
- Non-steady-state
 - Spatially varying meteorological fields
 - “Remembers” concentration from previous time-step (usually hour)

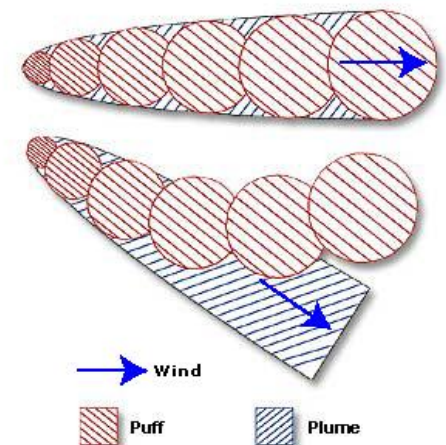
Steady-state Gaussian plume models

- Longest history in regulatory air quality modeling
- Use Gaussian plume models
- Assume homogeneous meteorological condition
- Example models
 - AERMOD
 - CALINE
 - ADMS-URBAN
 - RLINE



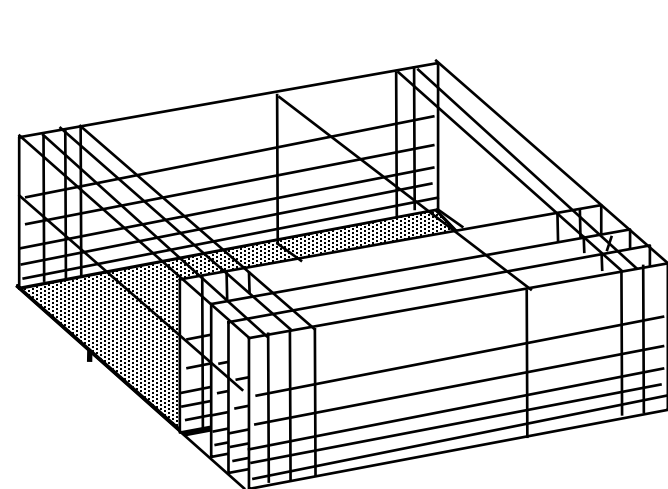
Non-steady state Gaussian models

- Pollutant modeled as “puff” or “particles”
- Movements of “puff” or “particles” are tracked spatially and temporally at each time step
 - Normally using 3-D meteorological field data from weather forecasting models
- Mainly applied in urban scale
 - Can be applied for regional scale
 - Long range transportation
- Examples: CALPUFF,
FLEXPART

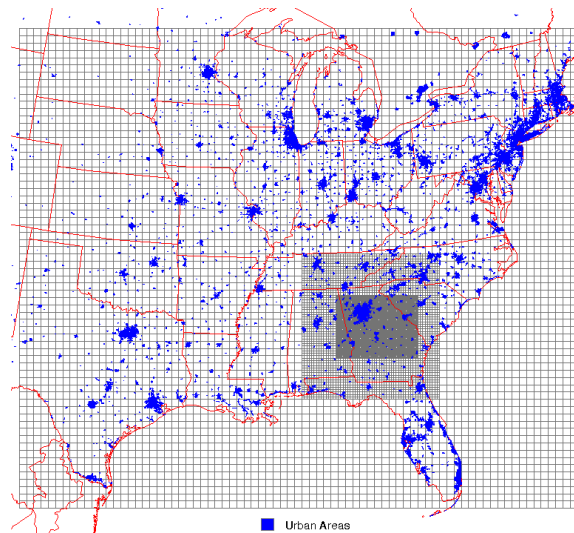


Chemical transport models

- Domain split into grids
- Setup conservation equations for each grid
 - Mass, energy, momentum
- All equations form a gigantic matrix, solved using algebra

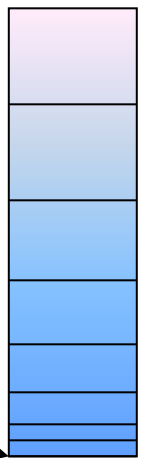


3D view



Spatial

About 15-30
vertical
Layers up to 15
km
(many in first 1 km)



Vertical

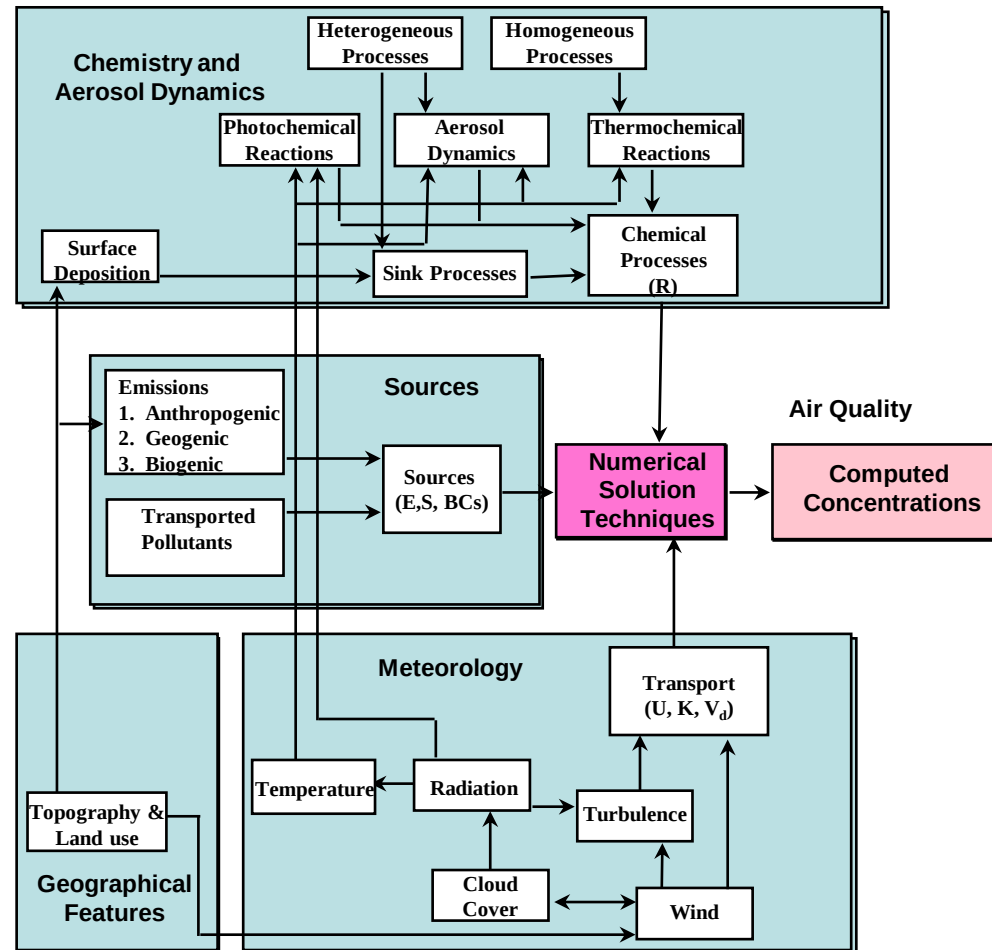
Eulerian chemical transport models

- Discretize the study area into grid cells
- Do not try to solve for meteorological fields
 - Data from weather forecasting models (e.g. WRF)
- At each time step
 - Solve for pollutant concentrations in each cell altogether from groups of mass balance equations
 - Homogeneous concentration distribution within cell
- Example: CMAQ, CAMx, UCD
- Generally applied in large scale, from regional scale (e.g. Eastern US) to global

Computational fluid dynamics models

- Virtually all use Eulerian approach
- Solve for meteorological field and concentrations at the same time
- Generally applied from continental to global scale
- Examples: GEOS-CHEM, WRF-CHEM, WRF-CMAQ etc

General framework of a modern air quality model



History of air quality models

- 1960s – 1980s: starting Gaussian, box, Lagrangian, photochemical mdls
 - E.g.: Empirical Kinetic Modeling Approach (EKMA) for O₃ modeling
- 1970s: Eulerian, aerosol, acid deposition mdls
 - E.g.: Urban Airshed Model (UAM) for photochemical
 - E.g.: Regional Acid Deposition Model (RADM)
- 1990s: EPA proposes “one atmosphere” concept, introduce Models-3
 - Community Modeling and Analysis System (CMAQ)

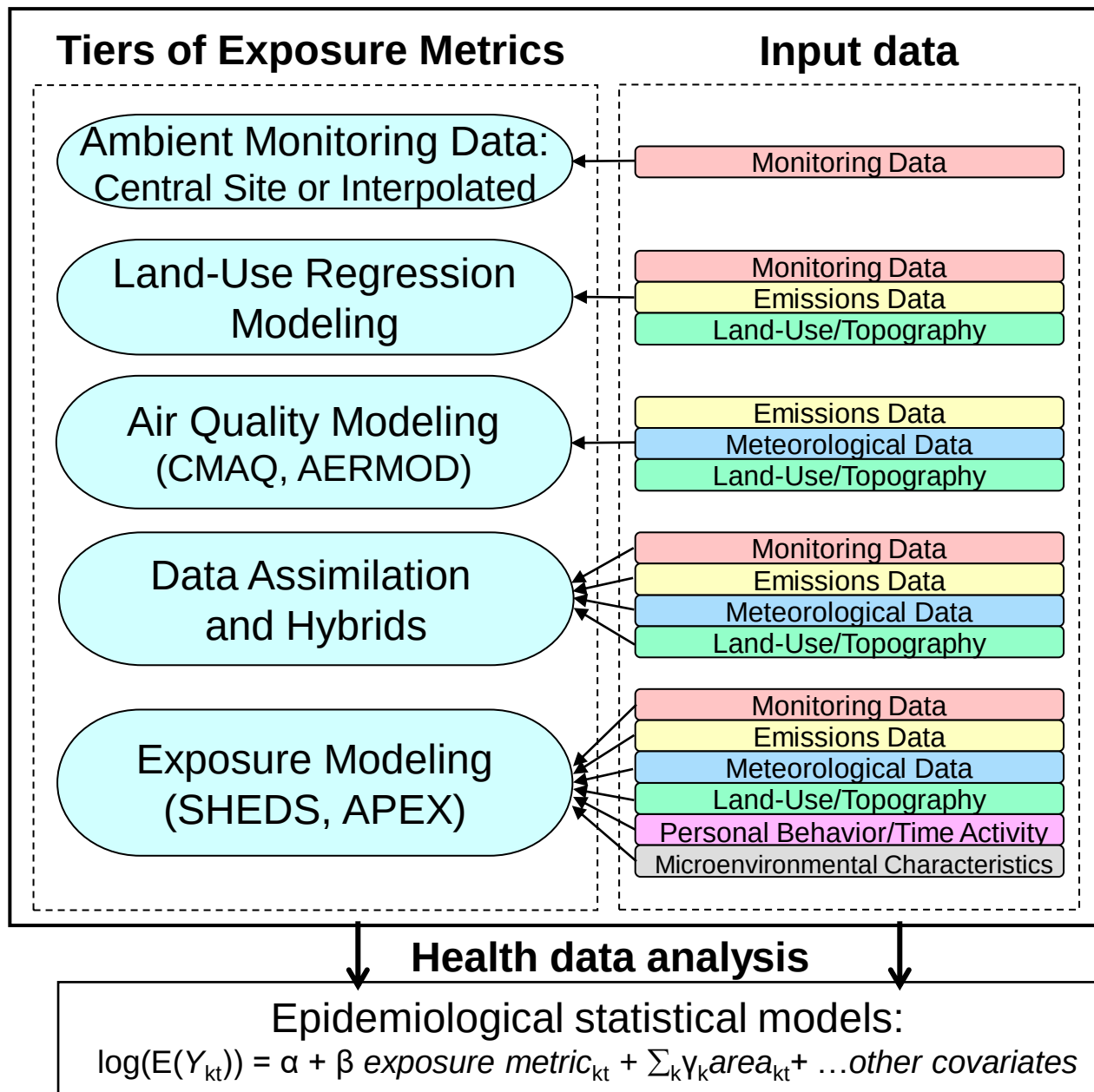
Air quality models in public health

Air quality models in public health

- Air quality models often used in health studies
- Mean to approximate spatial and/or temporal concentration distributions of focus pollutant(s)
 - No interest in actual physical chemical processes
- Use model to, mostly, quantify how much pollutants study subjects were exposed to
 - Exposure estimation

Exposure Information Relevant to Health Studies

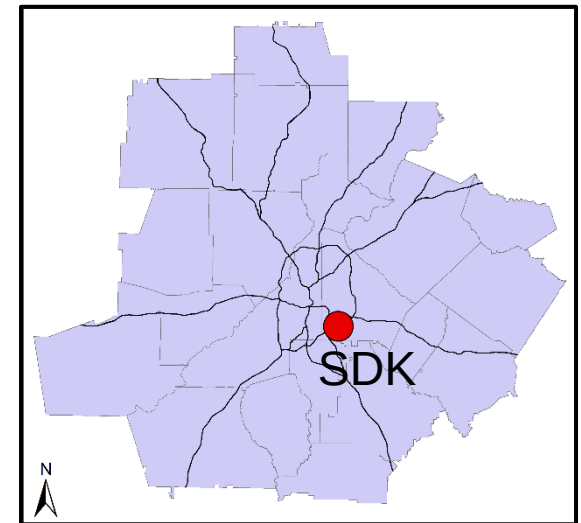
Complexity



Adapted from
EPA website

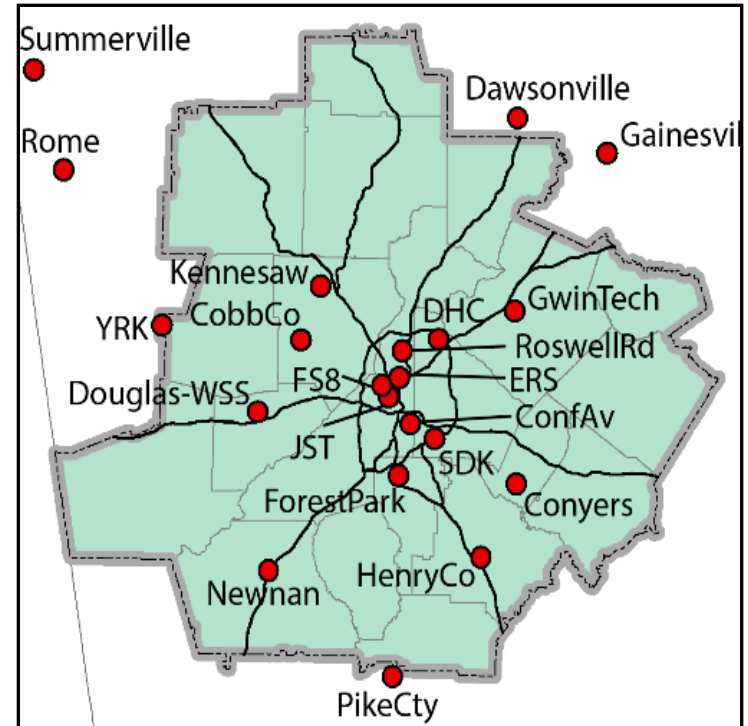
Central Monitor

- Uses data from a single measurement station
- Minimum data requirement easy to apply
- One monitor often do not represent a big area
- Spatially, temporally, chemically incomplete
 - E.g.: $\text{PM}_{2.5}$ components data often not available everyday



Site average

- Average monitor sites within a region
- Still spatially, temporally, chemically incomplete
 - Rely on available measurement data



Inverse Distance Weighting (IDW)

- Concentration at unknown location = weighted average of known concentrations

$$C_l = \frac{\sum_{i=1}^N w_{i,l} C_i}{\sum_{i=1}^N w_{i,l}} \text{ (when } l \text{ does not overlap with any monitor } i \text{)}$$

$$C_l = C_i \text{ (when } l \text{ overlaps with monitor } i \text{)}$$

$$w_{i,l} = \frac{1}{d(i,l)^2}$$

C_l is the interpolated concentration at location l ;

C_i is the measured concentration at monitor i ;

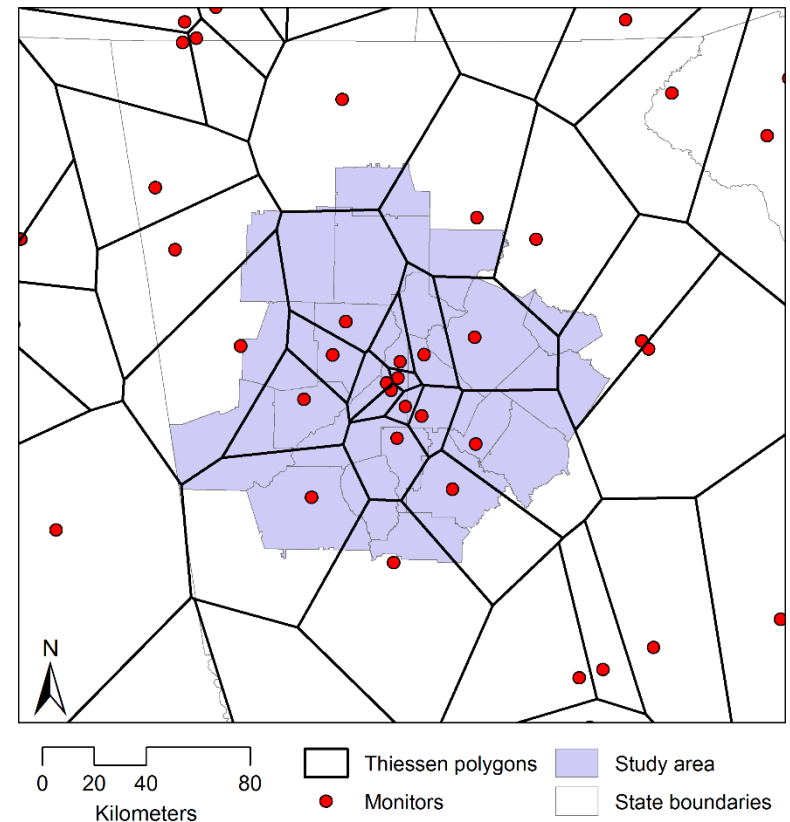
N is the total number of monitors with known concentration;

$w_{i,l}$ is the choice of distance weights between location l and monitor i .

$d(i,l)$ is the distance between interpolated location l and monitor location i .

Tessellation (TESS)

- Another spatial interpolation methods
- Split area into polygons
 - Each polygon contain one monitor
- Can assume inside polygon no spatial variations, or empirically estimate variability



Thiessen polygons

Kriging (KRIG)

- A more statistical rigorous spatial interpolation methods
- All kriging forms fit a parametric function (e.g. spherical, exponential, Gaussian, or quadratic) to the binned empirical semivariance as calculated by:

$$\gamma(d) = \frac{\sum_{i=1}^n [C(x_i + d) - C(x_i)]^2}{2n}$$

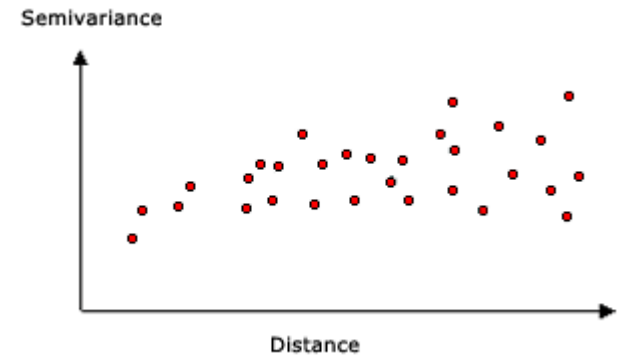
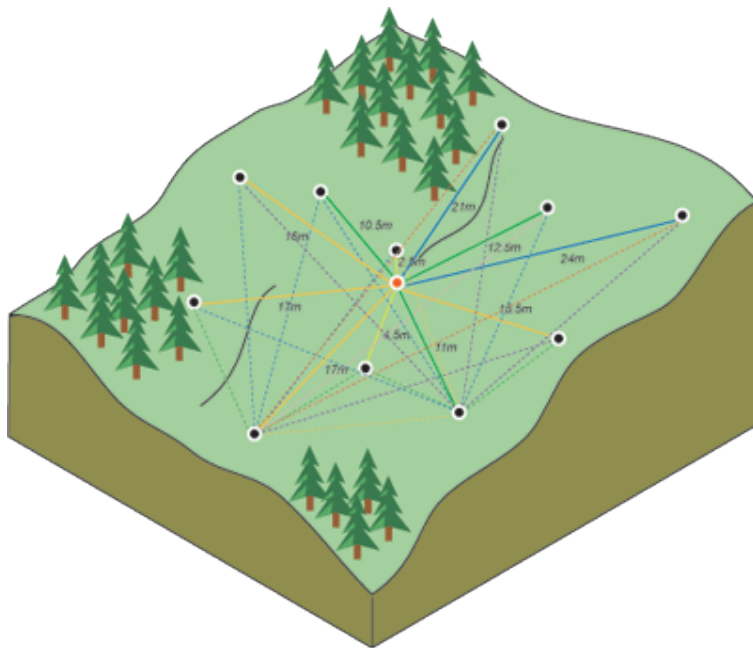
$\gamma(d)$ is the calculated semivariance for separation distance d ;

$C(x_i + d) - C(x_i)$ is the concentration difference between any pairs of monitors with separation distance d ;

n is the number of paired monitors available.

Kriging (KRIG)

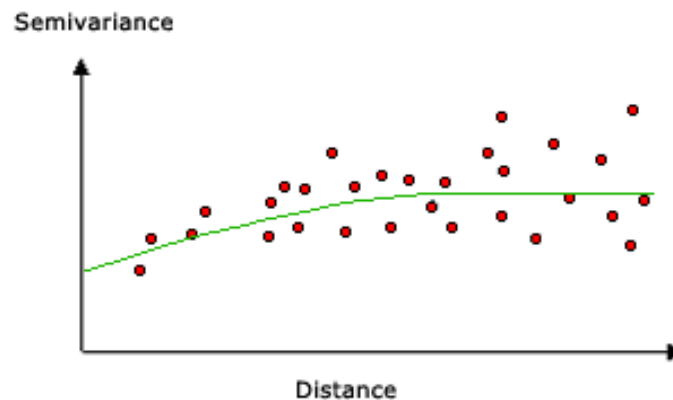
- Calculate “semivariances” between all pairs of points → semivariogram cloud



semivariogram cloud

Kriging (KRIG)

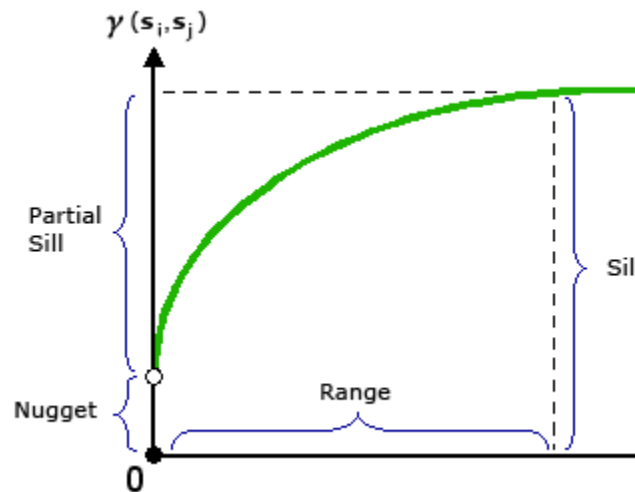
- Sort semivariogram cloud into distance bins
 - Often semivariogram cloud are very messy
- Fit a model on semivariogram, which can then be used to estimate concentration at unknown locations
- *Model provides uncertainty estimates*



Fit a spherical model

Kriging (KRIG)

- Fitted model has three key elements
 - Nugget: a non-zero semivariance value at distance of 0 (though the nugget is ideally 0).
 - Range: at this distance the model become flat
 - Sill: The biggest semivariance value



Land Use Regression (LUR)

- A empirical spatial interpolation method
- Relating observed concentrations to land use

$$Y(s_i, t_j) = \beta_0(s_i, t_j) + \sum_k \beta_{1,k}(s_i, t_j) X_k(s_i, t_j) + \epsilon(s_i, t_j)$$

$Y(s_i, t_j)$ is the observed concentration at location (monitor) s_i (where i is typically a monitor location) and time t_j ,

β_0 and $\beta_{1,k}$ are the regression coefficients

X are the independent variables (e.g., land use or meteorological parameters)

k is the index indicating type of land use

ϵ is the residual error term.

Land Use Regression (LUR)

- Example of developed LUR equation

$$C = 0.5 + 1.2 * L$$

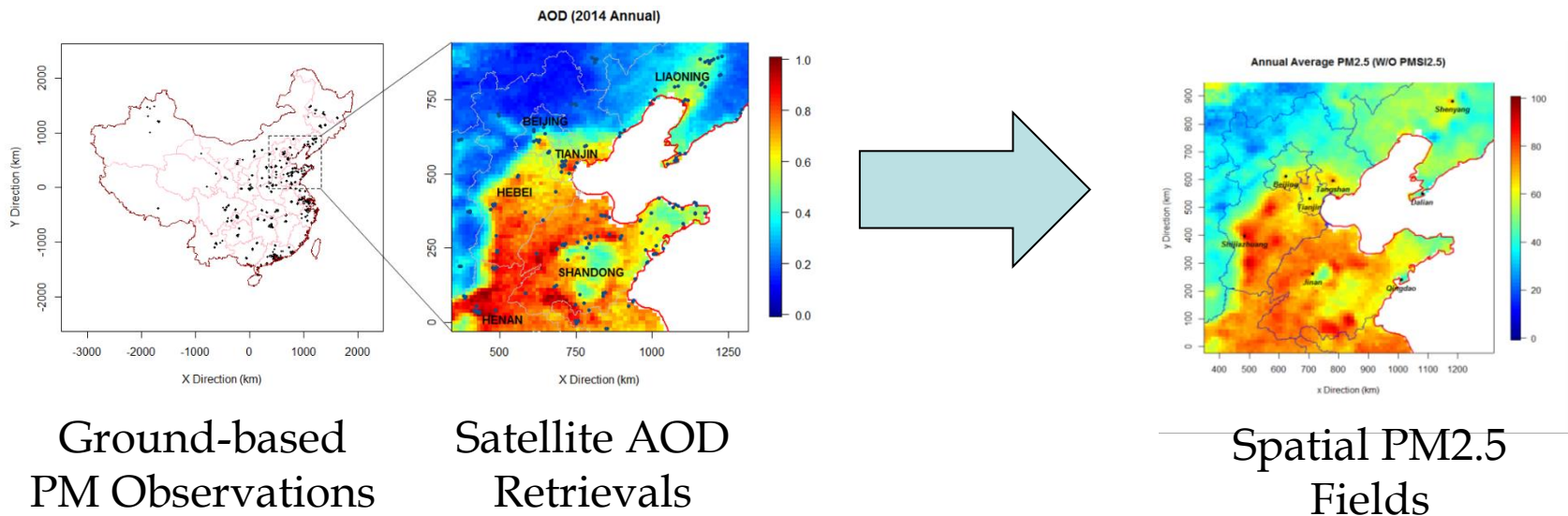
Meaning: PM concentration = 1.2 times the length of roadways (km) within 500 m distance to this location, and plus 0.5 $\mu\text{g}/\text{m}^3$

- Can also include many other independent variables such as meteorological (wind, T), emission data (power plants) etc
- Vary widely used in health association studies
- Equation developed in one location can't be used in another location

Aerosol Optical Depth (AOD)

- Similar with LUR, with AOD data added as one independent variable

$$Y(s_i, t_j) = \beta_0(s_i, t_j) + \beta_{1,1}(s_i, t_j)AOD(s_i, t_j) + \sum_k \beta_{1,k}(s_i, t_j)X_k(s_i, t_j) + \epsilon(s_i, t_j)$$



Roughly how AOD works: https://www.youtube.com/watch?v=YP0et8I_bvY

A video on AOD: <https://www.youtube.com/watch?v=aKbvcV3e5tc>

Aerosol Optical Depth (AOD)

- Global AOD coverage, some data in high resolution
 - 1 km resolution Multiangle Implementation Of Atmospheric Correction (MAIAC) data
- More and better data will become available
 - TEMPO (Tropospheric Emissions: Monitoring of Pollution)
- Missing retrieval is a major concern
 - Cloud interference etc